MIL-C-29109A(NU)
AMENDMENT-2
8 April 1987
SUPERSEDING
MIL-C-29109A(NU)
AMENDMENT-1
18 November 1983

MILITARY SPECIFICATION

COVERALLS, SUBMARINE DECK EXPOSURE

This amendment forms a part of Military Specification MIL-C-29109A(NU) dated 23 May 1978, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 1

Para 2.1 - Delete in its entirety and substitute the following:

" 2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation."

Delete last six lines and substitute the following:
"

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

"

Add under Federal Specifications - "NN-P-71 -Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)"

Page 2

Add under Military Specifications - "MIL-P-15011 - Pallets, Material Handling, Wood Post Construction, 4-Way Entry"

Add under Military Standards - "MIL-STD-147 - Palletized Unit Loads"

AMSC N/A FSC 8415

STATEMENT A. Approved for public release, distribution is unlimited."

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Para. 2.2 Following line 4 add: "AMERICAN SOCIETY FOR TESTING AND MATERIALS
D 1056-78 Flexible Cellular Materials-Sponge or Expanded Rubber

(Copies of ASTM Publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)"

Page 3

Add the following: "2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

Para. 3.3.2 line 2, after "conforming to" add "type III of"

Para. 3.3.4 line 2, delete "Class 1 Subclass B" and substitute "Class B". Add new sentence at end of paragraph: "For coverall lining, the color of the thread shall approximate the shade of the taffeta used."

Para. 3.3.5.1 line 2 Following "long," Delete the remainder of the paragraph and substitute the following "molded plastic individual element or continuous chain conforming to type I, style 3, size MS, short tab pull of V-F-106. The color of the plastic slide fastener components shall be black."

Page 4

Para 3.3.5.2 - Delete the first two sentences in their entirety and substitute the following: "The slide fastener shall be molded plastic individual element or continuous chain conforming to type I, style 15, size MHS, long tab pull of V-F-106. The color of the plastic slide fastener components shall be black".

Para. 3.3.5.2.1, line 1, Following "for" add "all". Delete "fastener" and substitute "fasteners".
line 3, Following "thong" add "for the front closure and utility pocket"
line 5, Following "inches" add "for front closure and 1-1/2 to 2 inches for utility pocket."

Add the following: "3.3.5.2.2 Sleeve closure The slide fastener shall be 7 inches long, molded plastic individual element or continuous chain, conforming to type I, style 3, size MS, short tab pull of V-F-106. The color of the plastic slide fastener components shall be black."

Para. 3.3.6 ,line 2 - following "straps" add ",and neck seal attachment"
line 3 - following "sleeve" add ",neck seal, neck tab,".

Para. 3.3.7, line 4 - Add "Ends of the cut cord shall be treated chemically with a clear resin finish to prevent unravelling."

Para. 3.3.8 label inscription, delete "DSA-100-00-0-0000" and insert "DLA-100-00-0-0000"
label inscription, delete last sentence of laundering instructions.

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3.3.10 - Delete in its entirety and add the following:

"3.3.10 Adhesive cement. The cement used for adhering butted seams of the foam interlining and neck seal and taping of butted joints of the foam interlining shall be a chloroprene compound. The use of reclaimed rubber is prohibited. The butted seams shall have a seam strength of 10 pounds per 2 inch width when tested as specified in 4.4.1.2.

3.3.10.1 Cemented seams. All butted seams of the foam interlining shall have a minimum breaking strength of 10 pounds per 2 inch width when tested as specified in 4.4.1.5.2.

3.3.10.2 Tape (cover). The tape used to cover all butted foam seams shall be 3/4 inch bias cut strips of the basic material (see 3.3.1).

Add the following - " 3.3.13 Seals 3.3.13.1 Chloroprene foam. The choloprene foam for the neck seal shall be pigmented black, gas expanded chlorinated cellular rubber, having a closed cell structure conforming to ASTM Classification RE-41-ET Test Method D-1056-78. The foam material shall be 1/8 (± 1/16) inch thick with skin on one side only (see 4.4.1).

3.3.13.2 Knitted stretch cloth. The stretch cloth shall be circular knit from 100% continuous filament nylon yarn and shall conform to the physical requirements below (see 4.4.1).

Knitted Stretch Cloth

Knit	Yarns per Inch (Min.) Wales	Courses	Weight Oz/yd^2	Percent Elongation Wales	Courses	Color
Jersey	36	38	2.4 ± 0.2	150 min 225 max	225 min 400 max	Black

3.3.13.3 Colorfastness. The knitted cloth shall show "good" fastness to laundering and perspiration equal to or better than the standard sample. When no standard sample is available the finished material shall show a minimum of "good" fastness to laundering, perspiration and a Munsell value for dry and wet crocking not lower than 8.5. (see 4.4.1).

3.3.13.4 Laminated foam. The knitted stretch cloth shall be laminated to the cut surface side of the chloroprene foam by means of a suitable adhesive, or other process capable of meeting the physical requirements below. The adhesive used shall be non-toxic, non-irritant and free from objectionable odors. The fabric backed (laminated) side shall be smooth and free of wrinkles or puckers and show no signs of blocking, tackiness or ahdesive strike-thru and shall conform to the requirements below, when tested as specified in 4.4.1.

Laminated foam

Characteristic	Requirement
Laundering resistance	No delamination
Adhesion, initial (lb/2 in)(min)	8.0
Adhesion, after laundering (lb/2 in)(min)	8.0
Adhesion, after aging (lb/2 in)(min)	8.0
Blocking	No blocking"

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Para. 3.4 - Following last line add: "The coveralls shall have a chloroprene foam neck seal, a slide fastener sleeve adjustment and a neck tab."

Para. 3.5.1 - Add the following to the pattern parts list:

"Cloth, coated, twill nylon (outershell)	Neck tab	2
	Sleeve placket	2"

Page 6

Para. 3.5.1 - Add the following to the pattern parts list:

"Chloroprene foam	Neck seal right	1
	Neck seal left	1
	Neck seal center	1"

Page 7

Table I, oper 1b, line 1 add at the beginning "Except for the drawcord channel,"

Table I, oper 1c, line 2, following "facing," add "sleeve plackets, neck tabs,".

Table I, oper 1f, line 1, following "plastic" add "or chloroprene"

Page 8

Table I, oper 4a, Delete in its entirety and substitute the following "4a. Cement all edges and butt all foam interlining to form the interlining shell, leaving sleeve seam open for slide fastener as indicated by pattern marks. Insure that all seams are properly cemented and are free of foreign matter prior to joining.".

Page 9

Add new operation 4c in its proper location: "4c. Cement and tape (see 3.3.10.2) all butted foam interlining seams. NOTE: Splices in the tape cover shall be permitted in long seams providing the splices are overlapped 3/4 inch and are no closer than two feet from each other".

Page 10

Table I, oper. 5i, line 2 - Delete "." and add the following "leaving seam open for slide fastener as indicated by pattern marks.".

Page 11

Table I, oper 5n, line 4 after "identification" add "label".

Page 12

Table I, Following oper. 6 add:

"Make sleeve plackets.

a. Fold sleeve plackets face to face lengthwise and stitch 1/4 inch from folded side and bottom edges. Turn placket to finished position and top stitch 1/16 inch from folded edges. 301 SSe-2 8-12 B B .

Attach slide fastener to placket/sleeve.

a. Position the right side of the slide fastener tape even with the raw edges of the sleeve placket and stitch 1/8 inch from the edge through all plies. Top stop of slide fastener shall finish 1/2 inch from the bottom finished edge of sleeve placket. 301 SSa-1 8-12 B B.

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b. Turn in seam allowance of lining even with the foam edges at sleeve openings. Lap the slide fastener tapes 1/4 inch over the foam edges and stitch 1/4 inch from back edges of slide fastener tape and 1/8 inch below the top stop. 301 LSaj-1 8-12 B B.

NOTE: The sleeve placket shall finish on front of the sleeve and the bottom edge of the placket shall be even with the finished sleeve bottom.

Page 13

Table I, oper. 7h, line 1 delete "tow" and "two".

Page 14

Table I, oper. 8b, line 3 after "wide" add "hook".

Table I, oper 9b, line 2 - Delete "." and add "for sizes small and medium and 7 inches for sizes large and x-large."

Page 20

Table I, oper. 19a, line 2 - Delete "." and add the following "leaving sleeve outseam open for slide fastener as indicated by pattern marks.".

Page 21

Table I, oper. 25c - Following the first sentence add: "Notch outershell seam allowance for sleeve opening as indicated by pattern marks and turn in seam allowance along both edges."

Page 22

Table I, Add the following in the proper sequence: "25e. Topstitch 3/8 to 1/2 inch from folded edge of sleeve openings, along both sides and 1/8 inch from the bottom stop. Stitching shall go thru all plies and secure the slide fastener tape and the top end of sleeve placket. 301 LSaj-1 8-12 B B"

Page 23

Table I, Add the following after operation 26.

"Make neck tab.

a. Position the neck tab pieces back to back with the pile fastener tape positioned and shaped according to pattern marks on the bottom ply and double stitch through all plies, with a 1/8 inch stitch gauge and the outside row of stitching 1/8 inch from the raw edges. 301 SSa-2 8-12 B B.

Attach neck tab and fastener tape.

a. Position neck tab on the left front according to pattern marks and in the direction to mate with the hook fastener tape on the right side of the neckline. Turn under raw edge of tab 1/4 inch and stitch to front. Folded edge of neck tab shall finish approximately 1/2 inch from finished front edge of the coverall. 301 LSbk-2 8-12 B B.

b. Center a 1 inch wide by 2 inch long piece of hook fastener tape on the neckline seam of the right front with the outside end of the tape 1/2 inch from the finished front edge and the long side parallel to the neckline seam. Stitch sides of the tape 1/16 to 1/8 inch from edge. 301 LSbj-1 8-12 B B.

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Assemble neck seal.

a. Cement (abut) the right and left neck seal pieces to the center neck seal piece. Insure that all seams to be cemented are free of foreign matter prior to joining. The skin side of the foam shall be considered the face.

b. Position a 4-1/2 inch long and a contour piece (according to pattern marks) of 2 inch wide hook fastener tape to the face side of the right neck seal in accordance with the marks on patterns. Stitch all sides of the tape 1/16 to 1/8 inch from the edge. 301 LSbj-1 8-12 B B

c. Cut a 1-1/2 inch long piece of 2 inch wide pile fastener tape shaped to fit the point of the left neck seal. Position that piece and a 4-1/2 inch long piece of 2 inch wide pile fastener tape to the back side of the left neck seal in accordance with pattern marks. Stitch all sides of tape 1/16 to 1/8 inch from edge. 301 LSbj-1 8-12 B B

Attach neck seal to the lining.

a. Position neck seal on the neckline of the lining assembly face to face matching marks on the top edge of the center neck seal with the lining neckline seam and cemented seams of neck seal with shoulder seams. Stitch seal to neckline in accordance with pattern marks to the lining only. 301 LSbk-2(a) 8-12 B B.

b. Line up inside edges of right and left neck seals with the inside edge of the facings and edge stitch to the facing and lining only. 301 LSbz-3(b) 8-12 B B

c. Stitch the right and left neck seals through all plies from the neckline to the bottom edge of the neck seal. Stitching shall be approximately 2-1/2 inches from the finished front edge of the coverall. 301 SSa-1 8-12 B B.

d. Position a 1 inch wide by 2 inch long piece of pile fastener tape to the inside of the right facing with the long outside edge parallel to and 1-1/2 inches from the front finished edge. The top end of the tape shall be approximately 2 inches from the finished neckline seam. Stitch all sides of the tape 1/16 to 1/8 inch from the edge thru all plies. 301 LSbj-1 8-12 B B.

e. Position a 1 inch wide by 2 inch long piece of hook fastener tape to the inside of the left facing with the long outside edge parallel to and 1-1/2 inches from the front finished edge. The top end of the tape shall be approximately 2 inches from the finished neckline seam. Stitch all sides of the tape 1/16 to 1/8 inch from the edge thru all plies. 301 LSbj-1 8-12 B B.

Table I, oper. 27 - Add the following: "c. All slide fasteners and hook and pile fastener tabs shall be in a closed position."

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Para 3.8, Table II - Delete measurements for size Large and X-Large sleeve inseams and substitute the following respectively: " 21 1/2 (54.6), 22 1/2 (57.1)"

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Add the following in the proper sequence: "4.1.2 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

Page 25

Para. 4.4.1, line 6 - Following "for" add "the foam requirements of 3.3.13.1, knit fabric requirements of 3.3.13.2,".

line 6 - Delete "3.3.5.1" and substitute "3.3.13.3"

Following last sentence add: "Component listed in Table III shall be tested for characteristics shown therein. The methods of testing as specified herein shall be followed. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The basis for lot size for laminated chloroprene foam shall be one sheet. The sample unit for testing shall be one square yard.

The sample size shall be in accordance with the following:

Lot Size	Sample Size
800 or less	2
801 thru 22,000	3
22,001 and over	5

Table III - Testing of components

Components	Characteristic	Determinations	Requirement Paragraph	Test Method
Laminated foam	Adhesive strike thru	1	3.3.13.4	Visual
	Wrinkled or puckered	1	3.3.13.4	Visual
	Laundering resistance	5	3.3.13.4	5556 1/
	Adhesion:			
	Initial	5	3.3.13.4	5950 2/
	After 3 launderings	5	3.3.13.4	5556 & 5950 2/
	After aging	5	3.3.13.4	4.4.1.6 & 5950 2/
	Blocking	3	3.3.13.4	5872 3/

1/ Laundering - Three cotton laundering cycles shall be performed on 8 by 8 inch specimens.

2/ Adhesion - Except that the distance between the clamps shall be two inches at the start of the test. Slitting the specimen between the nylon knit and foam to help separate the fabric from the foam sufficiently to fit the clamps is permissible. The adhesion shall be tested in the fabric's length (wale) direction.

3/ Blocking - Except specimens 4 by 4 inches shall be placed together fabric side to fabric side.

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In the proper sequence add the following paragraphs:

4.4.1.4 In-process inspection Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are carried out as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection had indicated non-conformance.

4.4.1.5 Intermediate inspection

4.4.1.5.1 Visual examination. Defects found during the intermediate examination shall be classified in accordance with 4.4.1.5.1.1. The applicable inspection levels and acceptable quality levels shall be as indicated in 4.4.1.5.1.3.

4.4.1.5.1.1 Examination of foam liner. The foam liner assembly shall be examined for defects in cutting, fitting and other construction characteristics which cannot be seen in the end item. The sample unit shall be one completely fabricated foam liner assembly. The lot size shall be expressed in the same terms as the sample.

	Classification		
Defect	Critical	Major	Minor
Any hole, cut or tear in the foam	X		
Construction not as specified		X	
Any component missing or other than the type specified		X	
Any component misplaced or not affixed as specified		X	
Any open seam	X		
Any seam not taped when specified	X		
(Butted seam) twisted, puckered or pleated			X
Thickness not as specified		X	

4.4.1.5.2 Break strengh of cemented seam For each 100 liners or less submitted for inspection, one completely fabricated foam liner shall be selected for seam test. The seam test shall be as described in paragraph 4.4.1.2 and the sample seams taken from the following areas: knee seam, outside arm seam, hood side seam, leg outseam, and leg inseam. Any test failure shall be cause for rejection of the lot.(see 3.3.10.1)

4.4.1.5.1.3 Foam liner inspection levels and Acceptable Quality Levels. The inspection levels and acceptable quality levels (AQL's) expressed as defect per 100 units for visual examination shall be as follows:

	AQL	Inspection Level
Major	2.5	II
Total (Major and Minor combined)	6.5	II

NOTE:If one or more critical defects are found during sampling inspection, the lot shall be rejected. For each lot found to be acceptable by the sampling plan, the balance of the lot shall be inspected for critical defects only. Any critical defect found shall be removed from the lot and replaced with a conforming item.

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Para. 4.4.2.1, Defect XVII c, line 1 - Delete "metal edge" and substitute "teeth".

Page 32

Para 4.4.2.1 - Add the following defects:

"XIX NECK SEAL

a. Seam not butted and cememted properly. i.e not smooth and flat - Major

b. Not stitched to coverall as specified. - Major

c. Hook and pile not positioned as specified, preventing a tight seal - Major

XX SLEEVE SLIDE FASTENER

a. Any part of the assembly omitted, bent, broken or otherwise defective -Major

b. Tape set on too loose or too tight, causing buldge, twist or fullness when closed - Major

c. Stitches too close to teeth not permitting slider to pass freely - Major

d. Top stop set more than 1/2 inch from finished bottom edge of sleeve placket - Major

e. Ends of slide fastener tape not securely caught in stitching - Major"

Para. 4.4.2.4 Delete in its entirety.

Page 33

Para. 4.4.3 - Following the last sentence add "1/ 1/For this defect, one item for each shipping container in sample shall be examined."

Add the following paragraph in the proper sequence"

4.4.4 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the end item inspected lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width,or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

Para. 5.2.1, line 3 - Following "type" delete the remainder of the sentence and the following sentence. Substitute the following "CF, class weather resistant, variety DW, grade V15c, size 2A of MIL-B-17757. The fiberboard for the liner shall conform to type CF, class domestic of MIL-B-17757.

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Para. 5.2.2, line 3 - Following "domestic," add "variety DW".
line 3 - Delete "200" and substitute "275".

Para 5.2.2.1 - Delete in its entirety.
Add the following in the proper sequence: "5.3.2 Labels, mixed sizes. Each shipping container, packed with mixed sizes only, shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 4 by 5 inches with the words "Mixed NSN'S" plainly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity and national stock numbers contained therein.

5.4 Palletization. When specified (see 6.2) item packed as specified in 5.2 shall be palletized on a 4-way entry pallet in accordance with load type 1a of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K, and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147.

The pallet shall be 4-way, Type IV; V Class 1, size; or Type VIII, fabricated from, wood groups I, II, III or IV, Grade A of NN-P-71, or 4-way, Style 1, Size A, Type I, Class 1 fabricated from wood groups specified, of MIL-P-15011. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer."

Para. 6.2, line 4 - Delete "." and add the following "the item tested should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, quantity, and testing and approval of the first article.
Lines 7-8 - Delete in their entirety and substitute the following:"e. When palletization is required. (see 5.4)"

Page 35

Add - "6.5 Subject term (key word) listing.
Chloroprene foam
Coated Nylon Twill
Coverall
Foam interlining
Submarine Deck Exposure"

Page 36
Figure 1 - Delete "Coverall" and substitute "Coveralls".

Custodian:
Navy - NU

Preparing Activity:
Navy - NU

User Activity:
Navy - SH

Review Activity:
DLA - CT

Project No. 8415-N369

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